

# CAC Controller (Standalone)

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## Installation Guide — Pioneer CAC CD Autochanger Controller

Detailed step-by-step guide for installing the CAC Controller on a Raspberry Pi. The controller operates a Pioneer CAC CD autochanger (V3000, V3200, V5000, or V180M) via a serial RS-232C connection.

This is the [CAC Controller \(Standalone\)](#) — the controller runs independently without Hub connectivity. For the Hub-capable version, see [cac-controller](#).

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# 1. Overview

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## What is the CAC Controller?

The CAC Controller is a Node.js web application that runs on a Raspberry Pi and controls a Pioneer CAC CD autochanger via the serial interface (RS-232C). It provides:

- [Responsive dark-theme UI](#) — Responsive dark-theme UI for browsers and smartphones
- [Control both players](#) — Control both players of the changer simultaneously
- [Catalog all CDs](#) — Catalog all CDs with covers, artist info, and track lists
- [Automatic metadata lookup](#) — Automatic metadata lookup
- [Automatically read and catalog CDs](#) — Automatically read and catalog CDs
- [Create custom playback lists](#) — Create custom playback lists
- [Installable as an app](#) — Installable as an app on smartphones

## Standalone vs. Hub-Capable Version

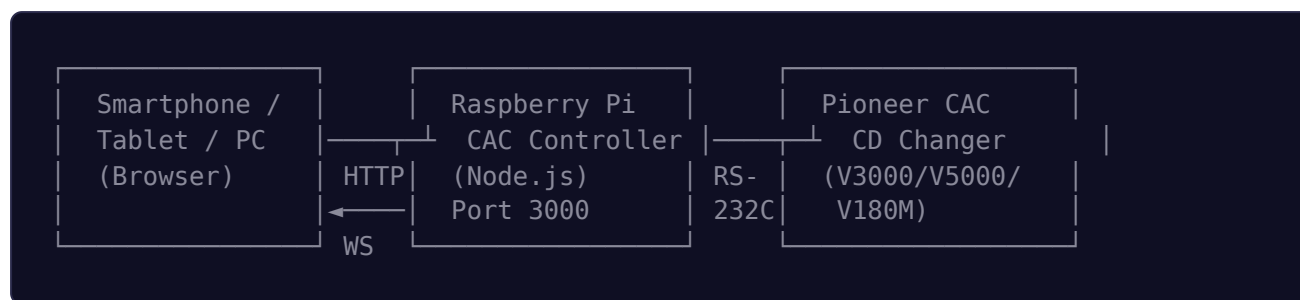
| Feature                                 | Standalone | Hub-Capable        |
|-----------------------------------------|------------|--------------------|
| Independent operation                   | Yes        | Yes                |
| All control functions                   | Yes        | Yes                |
| Hub connectivity (API key)              | No         | Yes                |
| Central management of multiple changers | No         | Yes (with CAC Hub) |

The Standalone version is ideal when only a single changer needs to be controlled and no central management is required.

## Supported Pioneer Models

| Model     | Slots | Baud Rate | Notes                                |
|-----------|-------|-----------|--------------------------------------|
| CAC-V3000 | 300   | 9600      | Standard reference model             |
| CAC-V3200 | 300   | 9600      | Same as V3000                        |
| CAC-V5000 | 500   | 9600      | 500-disc magazine                    |
| CAC-V180M | 18    | 4800      | Compact changer, different baud rate |

# System Architecture



## 2. Prerequisites

### Hardware

- **Raspberry Pi** — Recommended: Pi 3B+, Pi 4, Pi 5, or Zero 2 W
  - Pi Zero (first generation) works but is significantly slower during installation
  - Minimum 512 MB RAM
- **Storage** — Minimum 8 GB, recommended 16 GB or more
- **Power** — Appropriate USB power supply for your Pi model
- **Serial Port** — USB-A to RS-232C (DB9 or DB25)
  - Chipsets: FTDI FT232R, Prolific PL2303, or CH340 (all Linux-compatible)
  - Will be recognized as `/dev/ttyUSB0`
- **Serial Cable** — RS-232C cable to the Pioneer changer
  - 15-pin D-Sub connector (Pioneer side)
  - 9-pin D-Sub connector (adapter side) or 25-pin depending on adapter
  - Pinout: TX (Pin 2), RX (Pin 3), GND (Pin 5) — crossed (null modem)
- **Network** — WiFi or Ethernet
- **Power** — Powered on and operational

### Software

- **Operating System** — Lite or Desktop (Debian Bookworm or newer recommended)

- — Enabled (in Raspberry Pi Imager under settings)
- — For installing packages

## Network

- The Raspberry Pi must be on the same network as the control device (smartphone/PC)
  - Recommended: Static IP address for the Pi (see Section 3)
  - Default port: 3000 (configurable)
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## 3. Preparing the Raspberry Pi

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### 3.1 Installing the Operating System

1. Download : <https://www.raspberrypi.com/software/>
2. Flash Raspberry Pi OS Lite (64-bit) to the MicroSD card
3. In the Imager under (gear icon):
  - Set hostname (e.g., `cac-livingroom` )
  - Enable SSH
  - Set username and password (default: `pi` )
  - Configure WiFi (SSID and password)
  - Set locale/timezone

### 3.2 Connect via SSH

```
ssh pi@<IP_ADDRESS>
```

If the IP is unknown:

```
# From your local machine (right after first boot):  
ping cac-livingroom.local  
# or check your router's DHCP lease list
```

### 3.3 Update the System

```
sudo apt update && sudo apt upgrade -y
```

### 3.4 Configure Static IP Address (Recommended)

For Raspberry Pi OS Bookworm (NetworkManager):

```
# Show current connection:
nmcli connection show

# Set static IP (example for WiFi):
sudo nmcli connection modify "preconfigured" \
    ipv4.method manual \
    ipv4.addresses "192.168.1.100/24" \
    ipv4.gateway "192.168.1.1" \
    ipv4.dns "192.168.1.1"

# Restart connection:
sudo nmcli connection down "preconfigured"
sudo nmcli connection up "preconfigured"
```

Replace the connection name ( `preconfigured` ) with your actual name (see `nmcli connection show` ).

### 3.5 Disable WiFi Power Saving (Recommended)

Power saving mode can cause delays and connection drops:

```
# Check current status:
iw wlan0 get power_save

# Disable:
sudo iw wlan0 set power_save off
```

Make it permanent:

```
sudo tee /etc/NetworkManager/conf.d/wifi-powersave.conf > /dev/null << 'EOF'
[connection]
wifi.powersave = 2
EOF
sudo systemctl restart NetworkManager
```

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## 4. Installing Node.js

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### 4.1 Install Node.js 20 LTS

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install -y nodejs
```

### 4.2 Verify Installation

```
node --version  
# Expected output: v20.x.x (minimum v18.x)  
  
npm --version  
# Expected output: 10.x.x
```

### 4.3 Alternative: Node.js via nvm

If the NodeSource method causes issues:

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.0/install.sh | bash  
source ~/.bashrc  
nvm install 20  
nvm use 20
```

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## 5. Installing Build Tools

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The native SQLite extension ( `better-sqlite3` ) needs to be compiled:

```
sudo apt install -y build-essential python3
```

On a fresh Raspberry Pi OS, these packages are usually already present. This command ensures all required compilers and headers are installed.

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## 6. Setting Up the Serial Port

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### 6.1 Add User to dialout Group

So the CAC Controller can access the serial port:

```
sudo usermod -a -G dialout pi
```

Log out and back in for the group change to take effect:

```
# Either re-login:
exit
ssh pi@<IP_ADDRESS>

# Or just update the group:
newgrp dialout
```

Verify group membership:

```
groups
# Should include "dialout"
```

### 6.2 Connect and Check USB-Serial Adapter

1. Plug the USB-serial adapter into the Raspberry Pi
2. Check if it is recognized:

```
# List USB devices:
lsusb
# Should show the adapter, e.g.:
# Bus 001 Device 003: ID 0403:6001 Future Technology Devices International, Ltd FT232 Serial (UART)

# Check serial port:
ls -la /dev/ttyUSB*
# Expected output: /dev/ttyUSB0

# If not found, check kernel log:
dmesg | grep -i tty
```

## 6.3 Connect the Serial Cable

1. Connect the USB-serial adapter to the serial cable
2. Plug the serial cable into the  port on the Pioneer changer
3. Power on the Pioneer changer

## 6.4 Test Serial Connection (Optional)

Manually verify that communication works:

```
# Configure port (example for V3000 at 9600 baud):
stty -F /dev/ttyUSB0 9600 cs8 -cstopb -parenb

# Query status (sends "Player 1 Status" command):
echo -ne "1PS?X\r" > /dev/ttyUSB0

# Read response (wait 3 seconds):
timeout 3 cat /dev/ttyUSB0
# Should return a response like "1PS..."
```

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## 7. Installing the CAC Controller

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### Option A: From GitHub (Recommended)

```
cd /opt
sudo git clone https://github.com/NFDiJee/cac-controller-standalone.git cac-controller
sudo chown -R pi:pi /opt/cac-controller
cd /opt/cac-controller
npm install
```

The `npm install` process takes several minutes on a Raspberry Pi (up to 10–15 minutes on a Pi Zero), as `better-sqlite3` is compiled natively.

### Option B: Manual Copy

If Git is not available or preferred:



```
# On your local machine:
scp -r ./cac-controller/* pi@<PI_IP>:/opt/cac-controller/

# On the Raspberry Pi:
cd /opt/cac-controller
npm install
```

## Verify Installation

```
ls -la /opt/cac-controller/
# Should contain: server.js, package.json, src/, public/, ...

ls -la /opt/cac-controller/node_modules/
# Should show installed packages
```

---

## 8. First Start and Testing

### 8.1 Start the Server Manually

```
cd /opt/cac-controller
node server.js
```

Expected output:

```
[Server] Initializing database...
[Server] CAC Controller running on http://0.0.0.0:3000
[Server] Model: CAC-V3000
[Server] Serial: /dev/ttyUSB0 @ 9600 baud
```

### 8.2 Open in Browser

On your smartphone/PC, open in the browser:

```
http://<PI_IP>:3000
```

Example: `http://192.168.1.100:3000`

You should see the CAC Controller dashboard with:

- Player cards for Player 1 and Player 2
- Transport controls (Play, Stop, Pause, Next, Previous)
- Navigation (Library, Scanner, Playlists, Settings)

## 8.3 Test First Connection

1. Press `Enter` in the browser
2. The changer should load a CD and start playing
3. The player card should update with status (disc, track, time)

## 8.4 Stop the Server

Press `Ctrl+C` in the terminal to stop the server.

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## 9. Setting Up the Systemd Service

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A systemd service ensures the CAC Controller starts automatically on boot and restarts on crashes.

### 9.1 Copy Service File

```
sudo cp /opt/cac-controller/cac-controller.service /etc/systemd/system/
```

### 9.2 Adjust Service File (If Needed)

If your user is not `pi` :

```
sudo nano /etc/systemd/system/cac-controller.service
```

Contents of the service file:

```
[Unit]
Description=CAC Controller - Pioneer CD Autochanger Controller
After=network.target

[Service]
Type=simple
User=pi
Group=pi
WorkingDirectory=/opt/cac-controller
ExecStart=/usr/bin/node server.js
Restart=always
RestartSec=5
Environment=NODE_ENV=production

[Install]
WantedBy=multi-user.target
```

Change `User=` and `Group=` to the correct user.

## 9.3 Enable and Start the Service

```
# Reload systemd configuration:
sudo systemctl daemon-reload

# Enable service on boot:
sudo systemctl enable cac-controller

# Start service now:
sudo systemctl start cac-controller
```

## 9.4 Check Status

```
sudo systemctl status cac-controller
```

Expected output:

```
● cac-controller.service - CAC Controller - Pioneer CD Autochanger Controller
   Loaded: loaded (/etc/systemd/system/cac-controller.service; enabled)
   Active: active (running) since ...
```

## 9.5 View Logs

```
# Follow current logs live:
sudo journalctl -u cac-controller -f

# Last 50 lines:
sudo journalctl -u cac-controller --no-pager -n 50
```

## 9.6 Test Autostart

```
sudo reboot
```

After reboot (approx. 30–60 seconds), `http://<PI_IP>:3000` should be accessible again.

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## 10. Configuring Settings

In the browser under , the following settings can be adjusted:

### 10.1 Serial Port

| Setting   | Default                   | Description                                                          |
|-----------|---------------------------|----------------------------------------------------------------------|
| Port      | <code>/dev/ttyUSB0</code> | Serial port of the USB adapter                                       |
| Baud Rate | <code>9600</code>         | <code>9600</code> for V3000/V3200/V5000, <code>4800</code> for V180M |

### 10.2 Device Settings

| Setting   | Default                | Description                                                 |
|-----------|------------------------|-------------------------------------------------------------|
| Model     | <code>CAC-V3000</code> | Select Pioneer model                                        |
| Max Discs | <code>300</code>       | Number of slots: 300 (V3000/V3200), 500 (V5000), 18 (V180M) |

### 10.3 Network

| Setting | Default           | Description      |
|---------|-------------------|------------------|
| Port    | <code>3000</code> | HTTP server port |

|          |      |                               |
|----------|------|-------------------------------|
| Language | auto | Automatic, German, or English |
|----------|------|-------------------------------|

## 10.4 Apply Changes

After changes to port or serial settings, restart the service:

```
sudo systemctl restart cac-controller
```

## 11. Building the CD Database

After installation, the CD library is empty. There are three ways to populate it:

### 11.1 Automatic CD Scanning

The scanner has the changer load each CD, read the TOC (Table of Contents), and return the CD:

1. In the browser under
2. Enter (e.g., 1 )
3. Enter (e.g., 300 )
4. Click

Approx. 30–60 seconds per CD (load + read TOC + return). A full scan of 300 slots takes about 3–5 hours.

The scan can be paused at any time and resumed later. Already scanned CDs are preserved.

### 11.2 MusicBrainz Search

For individual CDs, metadata can be searched automatically:

1. In the browser under
2. Search by or
3. Select a result

4. Apply to a slot

## 11.3 JSON Import

If a database from a previous installation exists:

1. In the browser under
  2. Select a `cds.json` file
  3. Click
- 

## 12. Installing PWA on Smartphone

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The CAC Controller can be installed as a Progressive Web App (PWA) on your smartphone:

### iOS (Safari)

1. Open `http://<PI_IP>:3000` in Safari
2. Tap the  icon (square with arrow)
3. Select
4. Enter a name and tap

### Android (Chrome)

1. Open `http://<PI_IP>:3000` in Chrome
2. Tap the  icon
3. Select
4. Tap

The app then runs in full-screen mode like a native app.

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## 13. Updates

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## From GitHub Repository

```
cd /opt/cac-controller
git pull
npm install
sudo systemctl restart cac-controller
```

## Manual Update

1. Copy new files to the Pi (via SCP or USB)
2. Run `npm install` (if dependencies changed)
3. Restart service: `sudo systemctl restart cac-controller`

## Upgrading to the Hub-Capable Version

If you later want to use a Hub for central management, you can switch to the Hub-capable version:

1. Stop the service: `sudo systemctl stop cac-controller`
2. Back up the database: `cp /opt/cac-controller/data/cac.db ~/cac-backup.db`
3. Rename old directory: `sudo mv /opt/cac-controller /opt/cac-controller-standalone`
4. Install Hub-capable version (see its installation guide)
5. Copy database: `cp ~/cac-backup.db /opt/cac-controller/data/cac.db`
6. Start service: `sudo systemctl start cac-controller`

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## 14. Troubleshooting

### 14.1 Service Does Not Start

```
# Check logs:
sudo journalctl -u cac-controller --no-pager -n 50

# Service status:
sudo systemctl status cac-controller
```

```
# Start manually for detailed error output:  
cd /opt/cac-controller && node server.js
```

## 14.2 Serial Port Not Found

```
# List all USB devices:  
lsusb  
  
# List serial ports:  
ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null  
  
# Check kernel log:  
dmesg | grep -i tty  
  
# Is user in dialout group?  
groups
```

Unplug the adapter, wait 5 seconds, plug it back in. If `/dev/ttyUSB1` appears instead of `ttyUSB0` : change the port in settings.

## 14.3 No Connection to Changer

```
# Manually test serial connection:  
stty -F /dev/ttyUSB0 9600 cs8 -cstopb -parenb  
echo -ne "1PS?X\r" > /dev/ttyUSB0  
timeout 3 cat /dev/ttyUSB0
```

- Wrong cable (not a null modem cable)
- Wrong baud rate (4800 instead of 9600 or vice versa)
- Changer powered off or in standby
- TX/RX swapped

## 14.4 Port 3000 in Use

```
# Find process on port 3000:  
sudo lsof -i :3000  
  
# If an old process is running, stop the service:  
sudo systemctl stop cac-controller
```



Alternatively: choose a different port in settings (e.g., 8080).

## 14.5 npm install Fails

```
# Check build tools:
gcc --version
python3 --version

# If not installed:
sudo apt install -y build-essential python3

# Delete node_modules and reinstall:
rm -rf node_modules package-lock.json
npm install
```

## 14.6 WiFi Disconnects

```
# Check power saving mode:
iw wlan0 get power_save

# Disable (see Section 3.5)
sudo iw wlan0 set power_save off
```

## 14.7 Raspberry Pi Zero: Slow Installation

On a Pi Zero (first generation), `npm install` can take up to 15 minutes. Alternatives:

1. Compile on a more powerful machine and transfer `node_modules/` via SCP
- 2.

```
sudo dphys-swapfile swapoff
sudo nano /etc/dphys-swapfile
# CONF_SWAPSIZE=1024
sudo dphys-swapfile setup
sudo dphys-swapfile swapon
```

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## 15. Uninstallation

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## Remove Service

```
sudo systemctl stop cac-controller  
sudo systemctl disable cac-controller  
sudo rm /etc/systemd/system/cac-controller.service  
sudo systemctl daemon-reload
```

## Delete Files

```
sudo rm -rf /opt/cac-controller
```

## Remove Node.js (Optional)

```
sudo apt purge -y nodejs  
sudo rm -rf /usr/lib/node_modules
```